Advent of code : 2015 day 18

https://adventofcode.com/2015/day/18

In LaTeX

Many years later !

1 Read the input file

In this document, we define a « two dimensional » array to store the lights[1]. It defines two macros `\setpos` and `\getpos` that allow to respectively set and read the value at a given row and column in the array.

```
\def\setpos#1#2#3{\expandafter\xdef\csname LIGHT-#1-#2-\endcsname{#3}}
\def\getpos#1#2{\csname LIGHT-#1-#2-\endcsname}
```

To fill this array, we read the input file and treat each line one after the other. To do so, we use two counts `\idi` and `\idj` that will be the indices of the light in the array, and read the file line by line into `\content` Then for each line, we use a separate macro, `\treatLine{\content}` that will recursively treat each character of the line. If the character is a `.`, then nothing happens, and if it's another character (in the present case, it is always a `#`) then the corresponding case of the array is set to `1`. In these process, we manually increment the values of `\idi` and `\idj`.

We also define a macro `\size` which is the total size of the grid (`100` in the input, but `6` in the example).

```
% Open the file for reading
\newread\myfile
\openin\myfile=input.txt

\newcount\idi
\idi=0
\newcount\idj
\idj=0

% Define a macro to store the contents of the file
\newtoks\content

\continuetrue
\loop
\read\myfile to \content

\ifeof\myfile
    \continuefalse
\else
    \advance\idi by1
    \idj=0
    % \content
    \treatLine{\content}
    % \parseline{\content}
    % Line is \content
\fi
\ifcontinue
\repeat
```

[1] Source: https://tex.stackexchange.com/a/367490/243474.

```
% Close the file
\closein\myfile

\newif\ifcontinue
\makeatletter
% Fill the current line of the main array, from the input string
\def\@treatLine#1{%
    \advance\idj by1
    \ifx\relax#1%
        % End of the string
    \else
        \ifx #1.%
            % \the\idi\ \the\idj\ is Off
            \setbpos{\the\idi}{\the\idj}{0}     % For part II
            \else
            % \the\idi\ \the\idj\ is On
            \setpos{\the\idi}{\the\idj}{1}
            \setbpos{\the\idi}{\the\idj}{1}     % For part II
        \fi
        \expandafter\@treatLine
    \fi
}
```

2 Display the table, and count the lights

By making a nested loop, we can display the array and count the light that are on. Precisely, we use a count initially at 0 and use two `foreach` loops. Note that we need to use the `\global` keywords inside the loop[2].

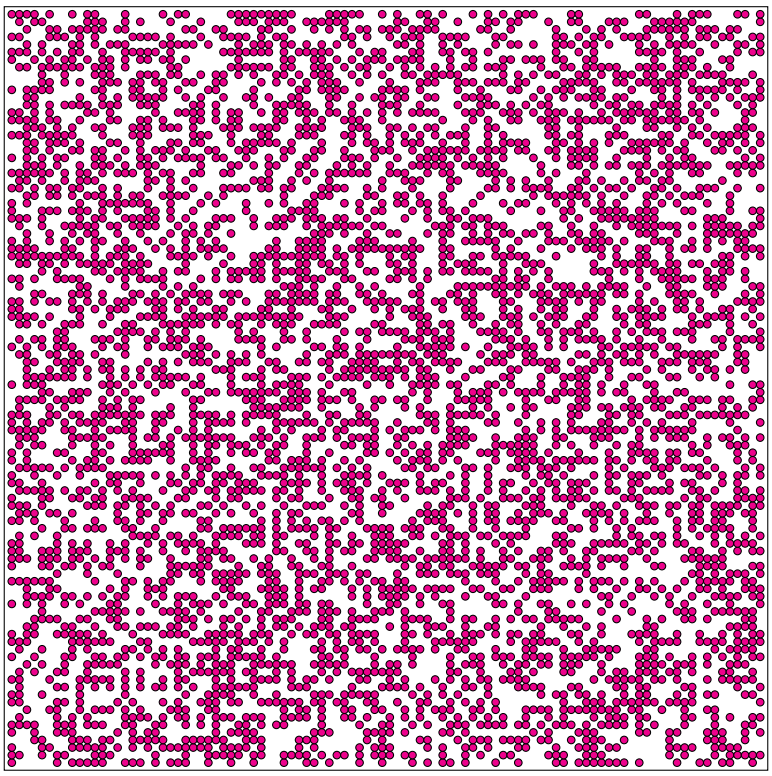
We define two macros `\display` and `\counton` that do that. Note the second is identical to the first on, but without the display using Ti*k*Z (so I do not rewrite the code here).

```
\newcount\nbon
\def\display{%
    \begin{center}
    \nbon=0
    \begin{tikzpicture}[scale=0.1]
        \draw (0, 0) -- (\size+1, 0) -- (\size+1, \size+1) -- (0, \size+1) -- cycle;
        \foreach \i in {1,2,...,\size}{%
            \foreach \j in {1,2,...,\size}{%
                \if\getpos{\i}{\j}1
                    \global\advance\nbon by1
                    \draw[fill=magenta] (\i, \j) circle[radius=0.5cm];
                \else
                    % To avoid excedding TeX capacity, we only draw on lights
                    % \draw[] (\i, \j) circle[radius=0.5cm];
                \fi
            }
        }
        \draw (\size/2, \size+1) node[anchor=south] {\the\nbon\ are On};
    \end{tikzpicture}
    \end{center}
}
```

Tiny remark : because of default axis orientation, there is a quarter-turn rotation in what is displayed here compared to what is on advent of code page. From here, apply a rotation of $-\frac{\pi}{2}$ to get what is displayed on the problem page.

[2] `https://tex.stackexchange.com/a/766457/243474`.

4905 are On



3 Before getting in the hard, some utils macros

In the following, we will apply the rule for switching on and off the lights. As to compute light at time step k we use the state at time $k-1$ that cannot be changed, we need to get a temporary array on which we will set the results, and in the end the values of this temporary array are applied back to the main light array.

First we define the macros `\setmpos` and `\getmpos` that are identical as the one for the light array, and we define the `\copyback` macro :

```
\def\copyback{%
    \foreach \i in {1,2,...,\size}{%
        \foreach \j in {1,2,...,\size}{%
            \global\setpos{\i}{\j}{\getmpos{\i}{\j}}
        }
    }
}
```

4 One step

To simulate one step of switching on and off the lights, we just apply the cellular automaton algorithm described on the page. This may not be the best way to do it, but I use define one count that count the number of light that are on. And other counts that contain the values of $i \pm 1$ and $j \pm 1$. These counters need to be defined outside the macro, this is due to TEX behavior, that will define a new one without erasing the previous ones, and eventually gets out of room to create new count.

```
\newcount\around
\newcount\ip
\newcount\im
\newcount\jp
\newcount\jm
```

```
\def\lightstep{%
    \foreach \i in {1,2,...,\size}{%
        \foreach \j in {1,2,...,\size}{%
            \around=0
            \ip=\i
            \advance\ip by1
            \im=\i
            \advance\im by-1
            \jp=\j
            \advance\jp by1
            \jm=\j
            \advance\jm by-1

            \if\getpos{\the\im}{\the\jm}1   % i-1, j-1
                \advance\around by1
            \fi
            \if\getpos{\the\im}{\j}1        % i-1, j
                \advance\around by1
            \fi
            \if\getpos{\the\im}{\the\jp}1   % i-1, j+1
                \advance\around by1
            \fi
            \if\getpos{\i}{\the\jm}1        % i, j-1
                \advance\around by1
            \fi
            \if\getpos{\i}{\the\jp}1        % i, j+1
                \advance\around by1
            \fi
            \if\getpos{\the\ip}{\the\jm}1   % i+1, j-1
                \advance\around by1
            \fi
            \if\getpos{\the\ip}{\j}1        % i+1, j
                \advance\around by1
            \fi
            \if\getpos{\the\ip}{\the\jp}1   % i+1, j+1
                \advance\around by1
            \fi

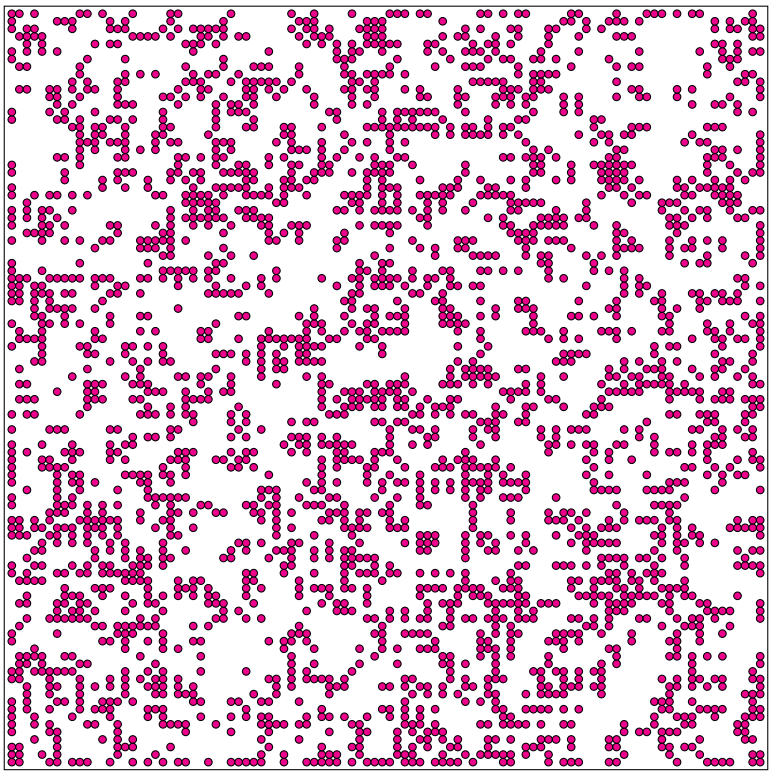
            \if\getpos{\i}{\j}1
                \if\the\around 2
                    \global\setmpos{\i}{\j}{1}      % Stays ON (2)
                \else\if\the\around 3
                    \global\setmpos{\i}{\j}{1}      % Stays ON (3)
                \else
                    \global\setmpos{\i}{\j}{0}      % Gets OFF
                \fi
                \fi
            \else
                \if\the\around 3
                    \global\setmpos{\i}{\j}{1}      % Gets ON !
                \else
                    \global\setmpos{\i}{\j}{0}      % Remains OFF
                \fi
            \fi
        }
    }
```

```
}
```

Then we just go on !

```
\lightstep
\copyback
```

2922 are On



5 100 steps

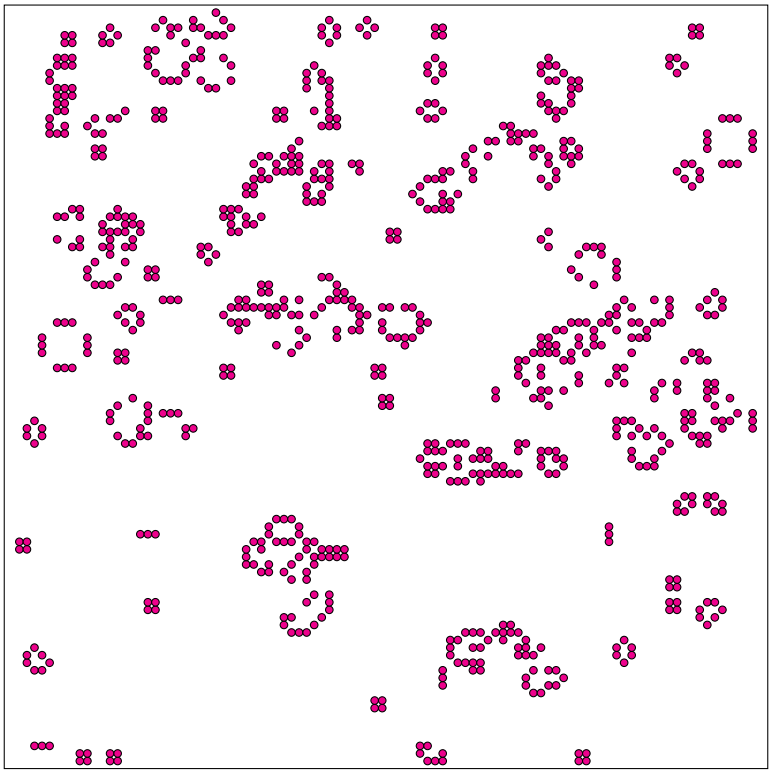
We just make a loop to call `\lightstep` and `\copyback` the desired number of time. As right above we already made one iteration, we begin the loop at 2. Here, I have commented the line to display, because with the huge grid the document would be very large.

```
\UseName{int_step_inline:nnn}{2}{100}{%
    \lightstep
    \copyback
    Iteration #1: \counton
    % \display
    \par
}
```

Iteration 2: 2683
Iteration 3: 2581
Iteration 4: 2438
Iteration 5: 2380
Iteration 6: 2242
Iteration 7: 2131
Iteration 8: 2177
Iteration 9: 2060
Iteration 10: 2019
Iteration 11: 1969
Iteration 12: 1931
Iteration 13: 1870
Iteration 14: 1904
Iteration 15: 1837
Iteration 16: 1792
Iteration 17: 1733
Iteration 18: 1698
Iteration 19: 1701
Iteration 20: 1672
Iteration 21: 1680
Iteration 22: 1621
Iteration 23: 1616
Iteration 24: 1606
Iteration 25: 1592
Iteration 26: 1535
Iteration 27: 1526
Iteration 28: 1576

Iteration 29: 1491
Iteration 30: 1504
Iteration 31: 1461
Iteration 32: 1416
Iteration 33: 1396
Iteration 34: 1423
Iteration 35: 1387
Iteration 36: 1387
Iteration 37: 1412
Iteration 38: 1466
Iteration 39: 1456
Iteration 40: 1481
Iteration 41: 1449
Iteration 42: 1497
Iteration 43: 1422
Iteration 44: 1446
Iteration 45: 1378
Iteration 46: 1427
Iteration 47: 1384
Iteration 48: 1399
Iteration 49: 1345
Iteration 50: 1312
Iteration 51: 1270
Iteration 52: 1249
Iteration 53: 1187
Iteration 54: 1222
Iteration 55: 1175
Iteration 56: 1153
Iteration 57: 1180
Iteration 58: 1074
Iteration 59: 1139
Iteration 60: 1075
Iteration 61: 1161
Iteration 62: 1096
Iteration 63: 1109
Iteration 64: 1075
Iteration 65: 1105
Iteration 66: 1082
Iteration 67: 1088
Iteration 68: 997
Iteration 69: 998
Iteration 70: 982
Iteration 71: 975
Iteration 72: 900
Iteration 73: 914
Iteration 74: 903
Iteration 75: 909
Iteration 76: 852
Iteration 77: 886
Iteration 78: 838
Iteration 79: 875
Iteration 80: 817
Iteration 81: 848
Iteration 82: 874
Iteration 83: 881
Iteration 84: 885
Iteration 85: 891
Iteration 86: 878
Iteration 87: 932
Iteration 88: 859
Iteration 89: 842
Iteration 90: 831
Iteration 91: 816
Iteration 92: 845
Iteration 93: 811
Iteration 94: 833
Iteration 95: 847
Iteration 96: 805
Iteration 97: 813
Iteration 98: 807
Iteration 99: 852
Iteration 100: 821

821 are On



Solution of part I : 821

6 Part II

Part II is the same as Part I, we run again the loop of 100 iterations, but at each time we need to slightly modify the configuration.

As we have made modifications to the main table, we need to create it back. Actually, in the macro `\treatLine`, we added two lines to fill a third backup array with the macro `\setbpos` with `0` or `1` whether the light is On of Off. And we also define another macro, `\reset` working the same way as `\copyback`, to put back the values of the backup array into the main one.

Then, we add a macro that manually make the four corner lights on, that will be called at each iteration of the loop :

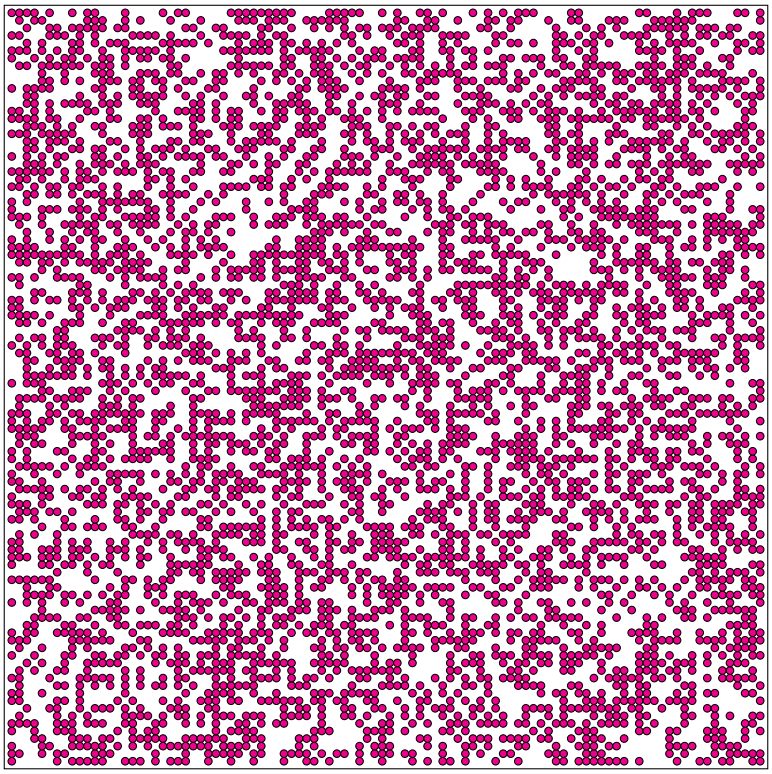
```
% The four corners lights are stuck on
\def\cornerLights{%
    \setpos{1}{1}{1}
    \setpos{1}{\size}{1}
    \setpos{\size}{1}{1}
    \setpos{\size}{\size}{1}
}
```

We set the thing up :

```
\reset
\cornerLights

\display
```

4906 are On



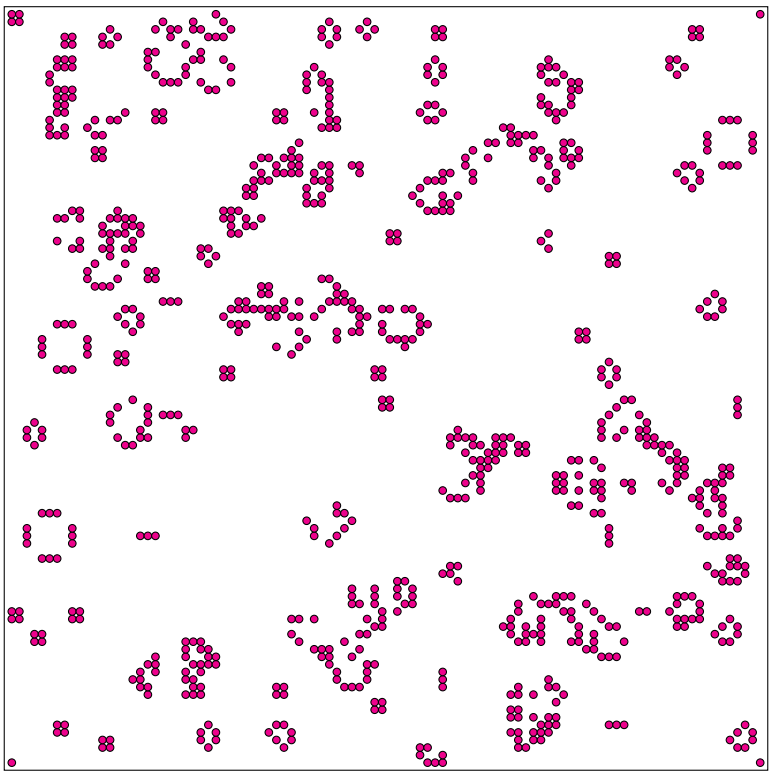
And we do the same loop as before (this time we begin at 1, because we have not made a first step), and we call `\cornerLights` at each iteration.

```
\UseName{int_step_inline:nnn}{1}{5}{%
    \lightstep
    \copyback
    \cornerLights
```

```
    Iteration #1: \counton
    % \display
    \par
}
```

Iteration 1: 2921
Iteration 2: 2680
Iteration 3: 2581
Iteration 4: 2439
Iteration 5: 2377
Iteration 6: 2244
Iteration 7: 2135
Iteration 8: 2181
Iteration 9: 2066
Iteration 10: 2022
Iteration 11: 1960
Iteration 12: 1921
Iteration 13: 1858
Iteration 14: 1886
Iteration 15: 1840
Iteration 16: 1771
Iteration 17: 1726
Iteration 18: 1687
Iteration 19: 1691
Iteration 20: 1673
Iteration 21: 1671
Iteration 22: 1631
Iteration 23: 1590
Iteration 24: 1552
Iteration 25: 1532
Iteration 26: 1462
Iteration 27: 1459
Iteration 28: 1502
Iteration 29: 1417
Iteration 30: 1400
Iteration 31: 1374
Iteration 32: 1342
Iteration 33: 1359
Iteration 34: 1401
Iteration 35: 1365
Iteration 36: 1339
Iteration 37: 1407
Iteration 38: 1454
Iteration 39: 1437
Iteration 40: 1469
Iteration 41: 1429
Iteration 42: 1474
Iteration 43: 1419
Iteration 44: 1443
Iteration 45: 1346
Iteration 46: 1377
Iteration 47: 1343
Iteration 48: 1317
Iteration 49: 1288
Iteration 50: 1262
Iteration 51: 1228
Iteration 52: 1203
Iteration 53: 1154
Iteration 54: 1181
Iteration 55: 1127
Iteration 56: 1111
Iteration 57: 1178
Iteration 58: 1062
Iteration 59: 1150
Iteration 60: 1065
Iteration 61: 1135
Iteration 62: 1074
Iteration 63: 1111
Iteration 64: 1071
Iteration 65: 1104
Iteration 66: 1077
Iteration 67: 1096
Iteration 68: 1017
Iteration 69: 992
Iteration 70: 975
Iteration 71: 949
Iteration 72: 887
Iteration 73: 920
Iteration 74: 886
Iteration 75: 905
Iteration 76: 867
Iteration 77: 894
Iteration 78: 861
Iteration 79: 868
Iteration 80: 812
Iteration 81: 822
Iteration 82: 826
Iteration 83: 814
Iteration 84: 847
Iteration 85: 854
Iteration 86: 847
Iteration 87: 895
Iteration 88: 852
Iteration 89: 878
Iteration 90: 823
Iteration 91: 846
Iteration 92: 864
Iteration 93: 845
Iteration 94: 865
Iteration 95: 859
Iteration 96: 856
Iteration 97: 875
Iteration 98: 866
Iteration 99: 907
Iteration 100: 886

886 are On



Solution of part II : 886

This file took 1 min 7.669 s to compile (without the call to `\display`, and 1 min 23.965 s with the call).